AI & ANALYTICS FOR BUSINESS OPTIMISATION

Assessment 1

AI and Analytics for Business Applications

BUSM2725

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1. Business Understanding

1.1 Primary Business Problem Definition

Modern Furniture Co. is a custom furniture manufacturer whose operational workflow moves sequentially through **Inquiry & Quoting**, **Design**, **Production** (Milling, Joinery, Finishing), and **Delivery & Reputation Management**. In this environment, customer inquiries arrive dynamically based on baseline demand, seasonal variations, and market reputation. Price quotes are algorithmically generated based on estimated labour hours and material costs.

The primary business challenge this report focuses on is **Demand Forecasting and Sales Quoting Optimization** under dynamic workshop capacity constraints. As demonstrated by comparing **Run A (Baseline: 40% Quote Win Chance)** and **Run B (Win60: 60% Quote Win Chance)** across a 730-day simulation:

- **The overbooking spiral:** Raising the quote win chance to 60% lifted the conversion rate from 39.4% to 49.5%. But the 15-person team could not handle the extra work. Cycle times rose to 19.1 days and only 57.3% of orders were delivered on time.
- **Reputation and sales:** Late deliveries damaged the company's reputation, which ended at 10.1% versus 84.5% in the baseline. Because reputation drives new inquiries, inquiries fell 40%, from 1,626 to 975.
- **Financial risk:** Late penalties reached $240,708.21. Cash ended at -$278,143.93, even though the company earned over $5 million in revenue.

This empirical evidence mirrors the findings of Shaik et al. (2026), who noted that unconstrained bidding on sales opportunities without operational capacity alignment causes severe margin erosion and suboptimal performance.

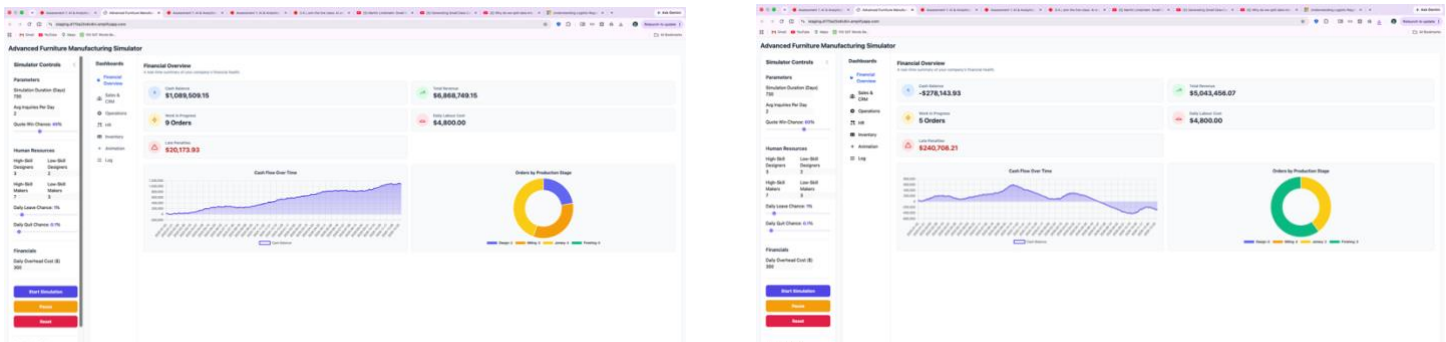


Figure 1 Simulator Financial Overview for (a) Run A - the baseline run (40% win chance) and (b)Run B - the Win60 (60% win chance) run. Baseline cash grows steadily to $1.09M. Win60 cash peaks near $0.6M in late 2025 and ends at -$278k. The difference in late penalties and revenue shows comparable difference in the two runs.



Figure 2 Business Problem Justification

1.2 Project Goal, Scope, and Research Question

Project Goal: Build and evaluate a machine learning model in Google Colab to forecast total customer inquiries over the next 7 days and use the results to guide quoting decisions that prevent workshop overload.

Project Scope: Analysis is bounded to the 730-day transactional, financial, and operational CSV datasets generated by the Advanced Furniture Manufacturing Simulator across Baseline and Win60 runs. Macroeconomic conditions outside the simulator's control panel are excluded.

Primary Research Question:

How can historical demand, workload and delivery data be used to forecast weekly inquiries and prevent workshop overbooking?

1.3 Business Need and Strategic Justification

Relying on manager intuition or static quoting targets in custom manufacturing leads to systemic overbooking or costly labor underutilization. Precise demand forecasting and capacity-aligned sales optimization address three critical business objectives:

1. **Financial Solvency & Cost Control**: Eliminates catastrophic late delivery penalties ($240.7k in Run B) and reduces unnecessary raw material holding costs.
2. **Reputation & Customer Protection**: Maintaining an On-Time Rate above 90% protects the 100% Reputation Score required to sustain organic lead generation.
3. **Capacity & Workforce Alignment**: Enables proactive workforce scheduling (balancing High-Skill vs. Low-Skill Designers and Makers). Quantitative forecasting using machine learning has been shown to reduce operating expenditures while improving sales revenue. (Mitra et al., 2022) (Shaik et al., 2026)

2. Data Acquisition and Understanding

2.1 Dataset Description & Key Variables

The analysis utilizes two core datasets exported directly from the simulator across both 730-day simulation runs (RunA - baseline and RunB - win60):

1. **All Orders (orders.csv)**: Captures order-level transaction details.
    - *Key Variables*: `id, productType, status , complexity, designHours, millingHours, joineryHours, finishingHours, materialType, materialQty, materialCost, quotePrice, creationDate, acceptanceDate, dueDate, completionDate, deliveryStatus, lateDays, penaltyPct.`
2. **Financial Ledger (ledger.csv)**: Captures daily cash flow transactions.
    - *Key Variables*: `date, day (1–730), type (COST vs. REVENUE), amount, description.`

2.2 Comparative Data Analysis (Baseline vs. Win60)

A high-level comparison of the two generated datasets demonstrates the impact of sales quoting parameters on workshop performance:

Table 1 Baseline (40%) versus Win60 (60%) results over 730 simulated days.

Metric	Run A: Baseline (40% Win Chance)	Run B: Win60 (60% Win Chance)	Operational Variance
Total Inquiries	1,626	975	**-40.0%** (Sales suppressed by reputation collapse)
Orders Won / Conversion Rate	640 (39.4%)	483 (49.5%)	+10.1% Win Rate, but lower overall volume
Reputation Score	**84.5%**	**10.1%**	**-74.4%** (Severe brand degradation)
On-Time Delivery Rate	~93.0%	**57.3%**	**-42.7%** (Massive delivery delays)
Average Cycle Time	13.5 Days	19.1 Days	+4.6 Days (Queue congestion)
Late Delivery Penalties	$0.00	**$240,708.21**	Incurred heavy cash penalties
Final Cash Balance	+$97,952.50	**-$278,143.93**	Severe net cash loss / insolvency

	Scenario	Total Inquiries	Orders Won	Conversion Rate (%)	On-Time Rate (%)	Avg Cycle Time (Days)	Total Penalties ($)	Final Cash Balance ($)
0	Baseline (40% Quote Win)	1626	640	39.36	93.98	13.08	403.48	1089509.15
1	High Demand (60% Quote Win)	975	483	49.54	57.32	19.09	4814.16	-278143.93

Figure3 Comparison of the Run A & Run B on Google Colab

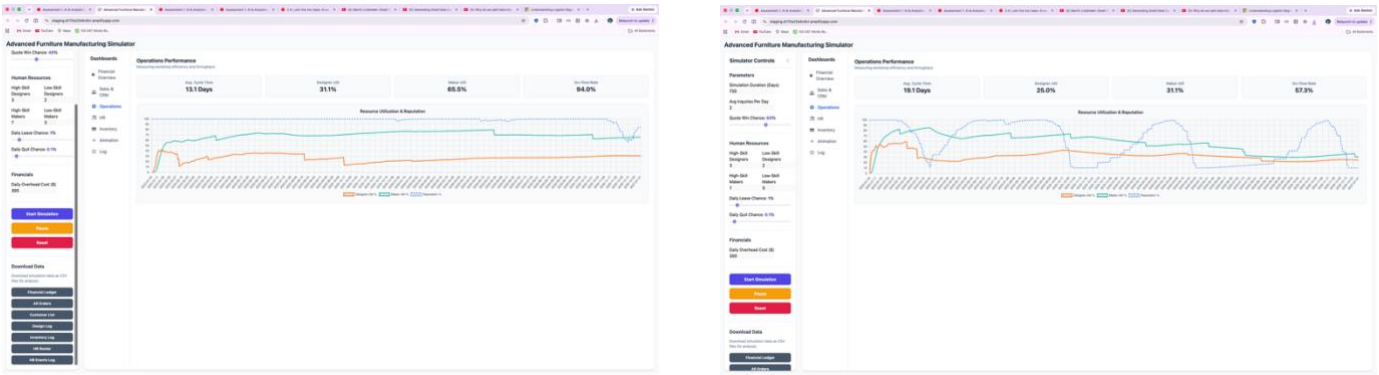


Figure 4 Simulator Operations dashboards for (a) the baseline run and (b) the Win60 run. In the baseline (40%) run the reputation consistently remains at ~90% whereas in the second simulation dashboard with quote win chance set as 60% - the reputation has serious fluctuations with steep lows.

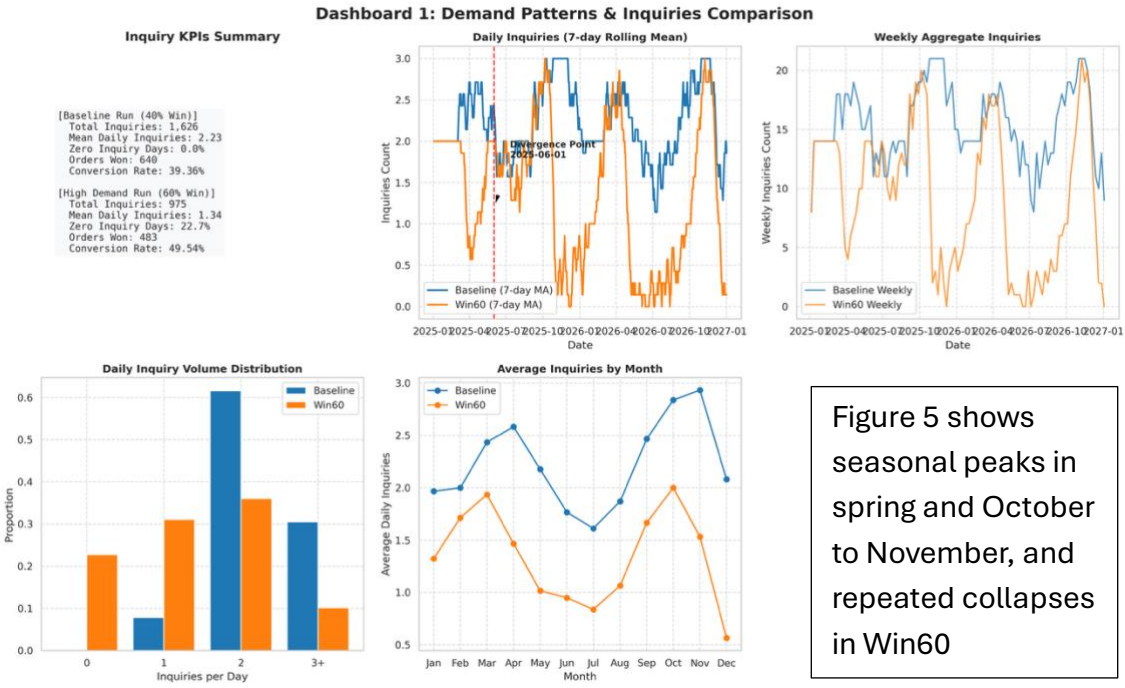


Figure 5 shows seasonal peaks in spring and October to November, and repeated collapses in Win60

Figure 5 Inquiry Patterns showcasing demand in both runs. (Visualisation from Google Colab)

2.3 Data Quality Assessment & Feature Engineering Challenges

Evaluating the generated dataset reveals several structural characteristics and data engineering requirements:

- **Data Quality & Completeness**:
  - For `LOST` quotes (~60% in baseline), fields like `completionDate`, `acceptanceDate`, and `dueDate` are null, which is expected behaviour for unconverted leads.
  - Timestamps are cleanly formatted (YYYY-MM-DD), allowing straightforward temporal aggregation.
- **Variable Relationships & Non-Linearities**:
  - Strong collinearity exists between complexity and required labour hours (`designHours`, `millingHours`, `joineryHours`, `finishingHours`).
  - An inverse non-linear relationship exists between `lateDays` and `reputation%`, creating a lagged feedback loop where late orders today suppress daily inquiry volume 10 to 30 days later.
- **Feature Engineering Plan**: To prepare the dataset for time-series forecasting and regression modelling in Google Colab, the following features must be constructed:

1. **Daily Aggregations**: Group order records by `creationDate` to compute `daily_inquiries`, `daily_orders_won`, and `daily_quote_value`.
2. **Rolling Lag Features**: Calculate 7-day and 14-day rolling averages for inquiry volume, win rate, and late delivery counts to capture short-term demand inertia. (Ferreira et al., 2016; Punia, 2025)
3. **WIP & Capacity Load Indicators**: Compute daily Work-in-Progress (`WIP_count`) and queue hours across Milling, Joinery, and Finishing stages to quantify shop-floor strain.
4. **Reputation & Price Ratio Features**: Calculate the moving average of the Reputation Score and ratio of `quotePrice` to `materialCost`.

3. Modelling

3.1 Approach and Rationale

I propose a regression model that forecasts total customer inquiries over the next 7 days. Regression suits the question because management needs a number, not a yes/no. A weekly total is also steadier than daily inquiries (only 0 to 3 a day) and gives a week's notice to adjust quoting and staffing.

Three algorithms are compared against a benchmark:

- Ridge Regression (simple and easy to explain) *(Li et al., 2023)*
- Random Forest (captures non-linear patterns and copes with small datasets) *(Breiman, 2001)*
- XGBoost (boosted trees, strong on tabular data). *(Chen & Guestrin, 2016)*

The benchmark is a naive forecast that repeats the last 7 days of inquiries. A model is only worth using if it beats this.

3.2 Data Preparation, Chronological Splitting, and Safety Gap

- **Data Preparation & Feature Engineering**: Transactional order logs (orders.csv) and financial ledgers (ledger.csv) from both 730-day simulation runs (**Baseline: 40% Quote Win** and **Win60: 60% Quote Win**) were merged into a zero-filled daily calendar table. A 30-day burn-in window was applied to construct rolling lag features (e.g., `inquiries_7d`, `inquiries_14d`, `orders_won_7d`, `wip`, `wip_hours`, `on_time_rate_14d`, `late_penalties_7d`), yielding **693 clean daily observations per scenario** (1,386 total).
- **Chronological Train/Test Split**: A chronological split with an embedded safety gap prevents temporal data leakage, following the Microsoft Team Data Science Process lifecycle (Microsoft, 2024) and time-series validation best practices (Punia, 2025). Therefore, an **80% Train / 20% Test chronological split** was performed within each run.

- **Enforcing a 7-Day Safety Gap**: Because the target variable (`target_7d`) aggregates customer inquiries over the next 7 days, a strict **7-day chronological gap (2026-08-09 to 2026-08-15)** was placed between the ***training period (2025-02-01 to 2026-08-08; 554 days)*** and ***test period (2026-08-16 to 2026-12-25; 132 days).*** This guarantees that no forward-looking target information from training days leaks into the feature windows of test observations.
- **Time-Series Cross-Validation**: Hyperparameters were tuned using an expanding-window time-series cross-validation scheme (4 folds) with an embedded 7-day safety gap to ensure realistic out-of-sample optimization.

3.3 Performance Evaluation Metrics

Data were split by time- because shuffling lets the model learn from the future.

- In each run, the first 80% (Feb 2025 to Aug 2026) trains the model, a 7-day gap follows so forecast windows do not overlap, and the last 132 days (Aug to Dec 2026) form a test set used once.
- Settings were tuned by grid search with time-series cross-validation (four expanding folds, same gap) scored on MAE.
- Performance is measured with MAE (average miss in inquiries), RMSE (penalises large misses), R-squared and WAPE (error as a percentage of volume), and compared with the naive benchmark.
- Permutation importance and ablation tests then show which feature groups drive accuracy.

Table 2 Test-set forecast error by model and run

Model	Baseline MAE	Baseline WAPE (%)	Win60 MAE	Win60 WAPE (%)	Win60 gain vs naive
Naive 7d	1.561	9.30	2.152	21.40	0%
Naive 30d	2.495	14.86	4.595	45.71	-114%
Ridge	1.602	9.54	1.788	17.79	+16.9%
Random Forest	1.700	10.13	1.506	14.98	+30.0%
XGBoost	1.622	9.66	1.608	16.00	+25.3%

4. Analysis and Recommendations

4.1 Key Insights

1. **Forecast value depends on conditions.** In the baseline run the naive forecast was best. In the 60% run, Random Forest cut error by 30% (Table 2, Figure 4).

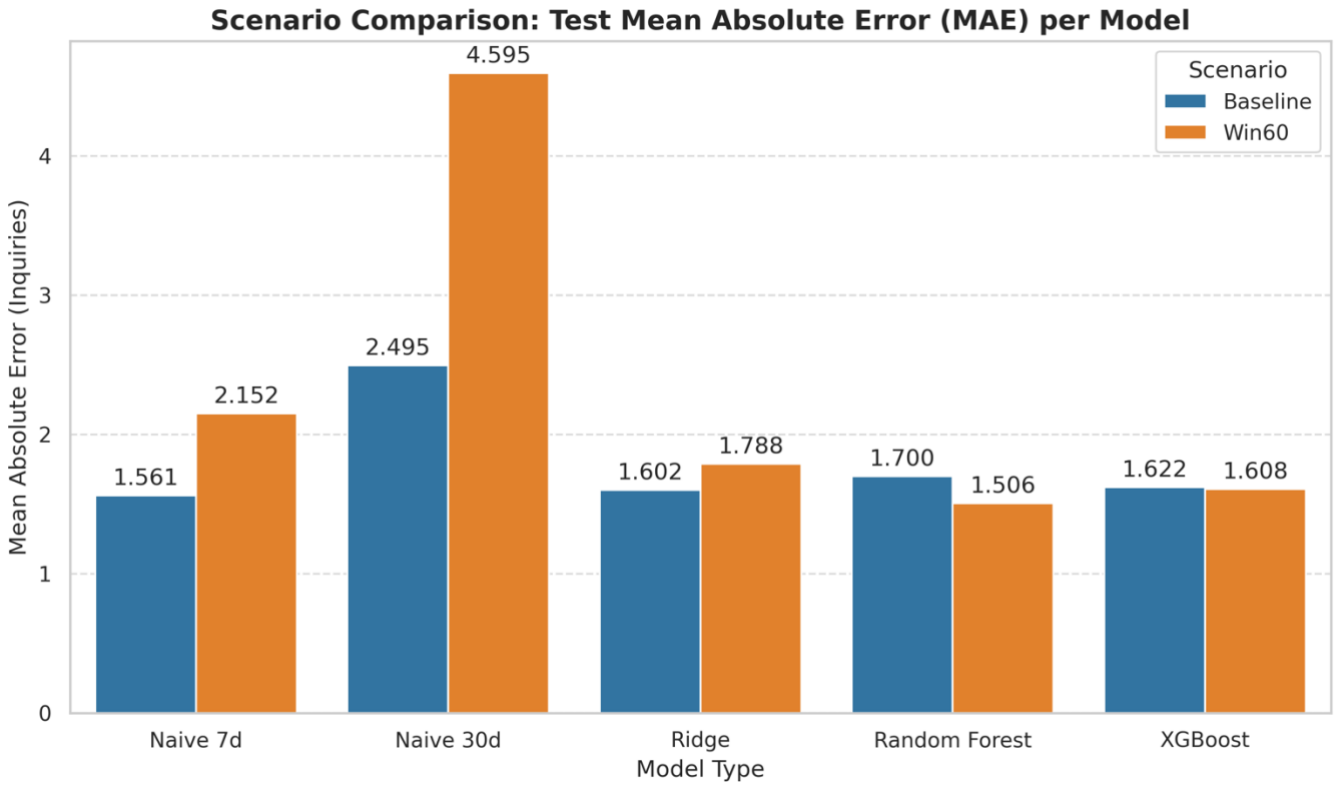


Figure 6 Test MAE by model and run. (Visualisation from Google Colab)

The naive 7-day forecast is best in the baseline run and Random Forest is best in Win60.

2. **Overbooking backfires.** Raising quote win chance from 40% to 60% lifted conversion (39.4% to 49.5%) but overloaded the workshop. On-time delivery fell from 94.0% to 57.3%, late penalties rose from $20,174 to $240,708, inquiries fell from 1,626 to 975, and closing cash dropped from about $1.09M to -$0.28M.

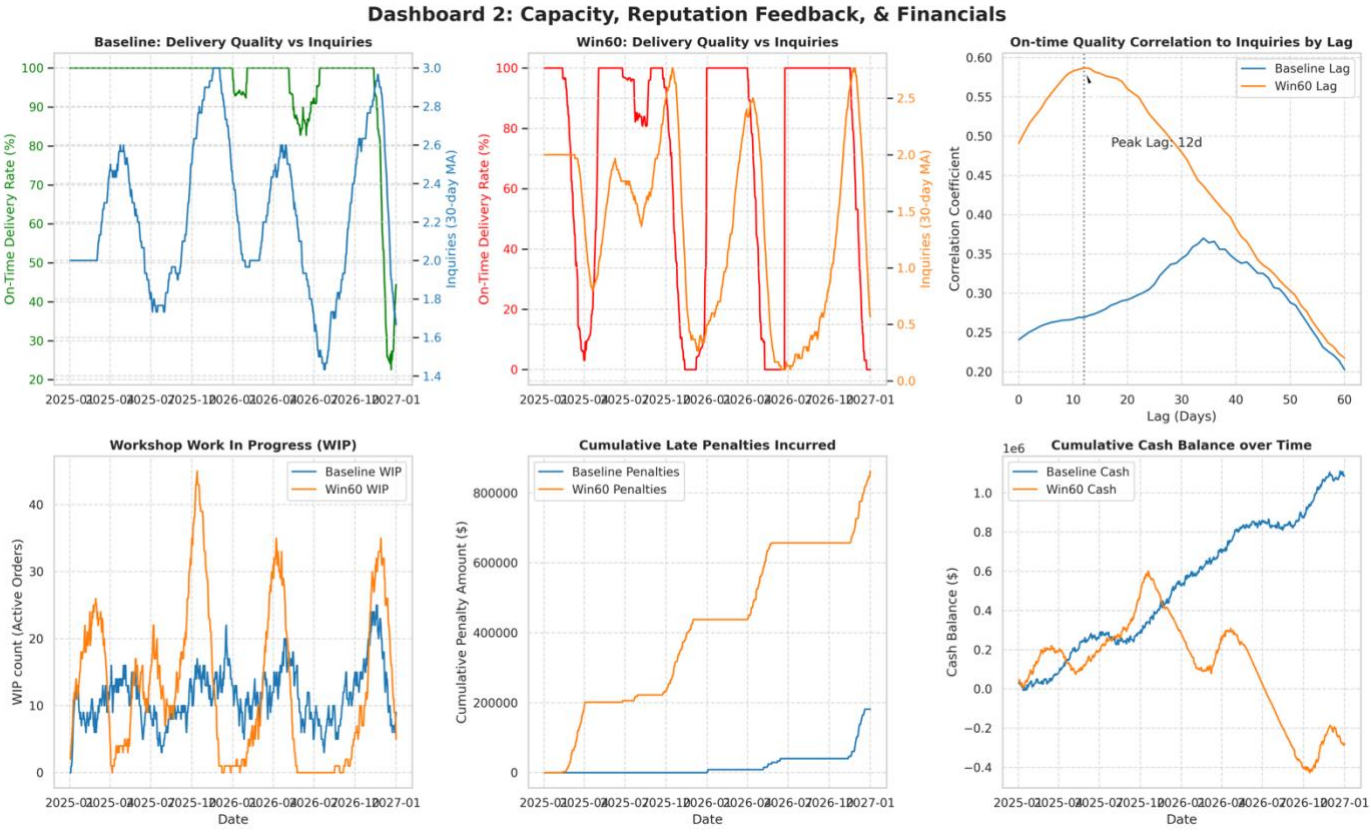


Figure 7 Dashboard showing workshop overload in 60% Run. (Visualisation from Google Colab)

3. **Recent delivery performance is the useful signal.** In the 60% run, removing on-time features raised Random Forest MAE from 1.51 to 2.25, no better than naive. Penalty, cash and work-in-progress features added nothing measurable. This fits the reputation loop, where late deliveries today mean fewer inquiries later, though it shows association, not proof.

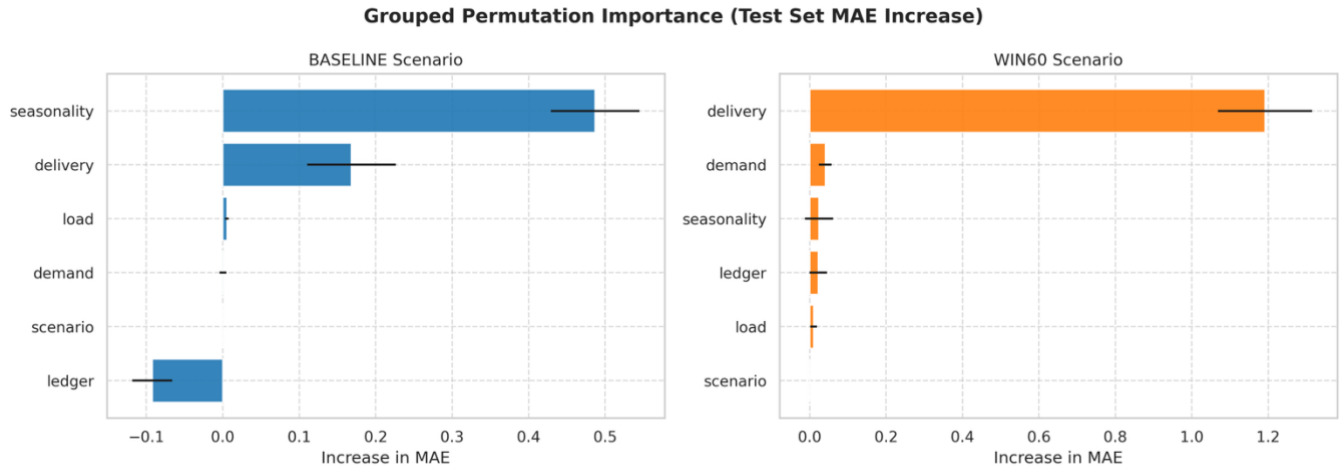


Figure 8 Increase in test MAE when each feature group is analysed. Delivery feature dominates in Win60.

(Visualisation from Google Colab)

4. **Models help when demand shifts.** On days with sharp changes, models roughly halved the naive error (60% run: 3.96 to 1.46 for XGBoost). On stable days, naive was better. These days were identified with hindsight, so this is descriptive.

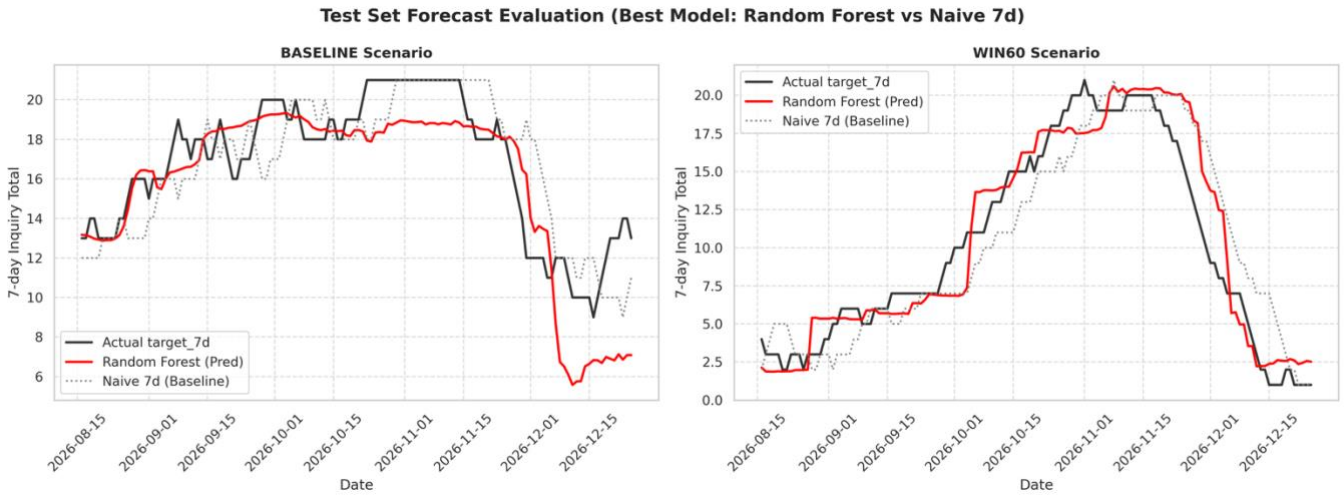


Figure 8 Random Forest forecast, naive forecast and actual 7-day inquiries on the test period. The model trails sharp turns.
(Visualisation from Google Colab)

5. **The model forecasts demand but cannot choose the best win chance,** as quote win chance had almost no importance.

4.2 Recommendations

- **Do not raise quote win chance beyond what the workshop can deliver.** Stay near 40% unless capacity is added (insight 2).
- **Track the rolling 30-day on-time rate as an early warning.** When it slips, expect weaker demand and slow quoting (insight 3).
- **Use the simple last-7-days forecast when deliveries are on time and demand is steady,** and switch to the model when on-time performance slips (insights 1 and 4).
- **Before accepting orders, compare work in progress and forecast inquiries with workshop hours,** and lengthen quoted lead times when the shop is full (insight 2).

4.3 Applications and Future Research

Practical applications

- **Manager dashboard:** Show the 7-day inquiry forecast in a simple dashboard (e.g. Power BI or Streamlit). Warn managers when work in progress nears workshop capacity, so they can slow down quoting.
- **Capacity planning:** Use the forecast to plan staff hours ahead of the October to November peak.
- **Material ordering:** Use the forecast to time timber reorders (Tasmanian Oak, Victorian Ash, American Walnut) and avoid stockouts and excess stock. This is a lower priority, as stockouts were rare in the simulation.

Future research

- **Model plus manager judgement:** Combine the forecast with managers' expert adjustments (a Demand Forecasting Decision Model (DFDM) with a step that corrects bias), so people and model check each other. *(Punia, 2025)*
- **Advanced models:** Test hybrid deep learning models (e.g. LightGBM-LSTM or ELM-HHO) once more data is available. *(Chaudhuri & Alkan, 2022) (Falatouri et al., 2022)*
- **Order optimisation:** Pair forecasts with integer linear programming, a maths method that finds the best mix of orders within capacity, to choose which orders to accept, their pricing and their scheduling together. (*Li et al., 2023)*
- **Wider testing:** Try more win-chance levels (e.g. 25% and 50%) with repeated runs, test a 14-day horizon, and validate on real data, since the simulator caps demand at 3 a day.

4.4 Limitations and Risks

Limitations

- **Small evidence base:** the results come from two simulated runs and one 132-day test window each, so they are indicative, not proof.
- **Simulated data:** the simulator caps demand at 3 inquiries a day and does not export reputation, so real-world results may differ.
- **Association, not cause:** the link between late deliveries and fewer inquiries is a pattern in the data, not a proven cause.

Ethical and business risks

- **Over-reliance:** the model reacts to turns rather than predicting them and gave a false alarm in the baseline run. Managers should treat it as decision support, not an automatic decision-maker.
- **Transparency:** managers should be able to see why a forecast changed (for example, that on-time delivery has slipped). This is one reason for choosing simple, explainable models.
- **Fair treatment:** forecasts should not be used to push staff into overtime or to unfairly turn customers away. Workload and delivery promises need to stay realistic.

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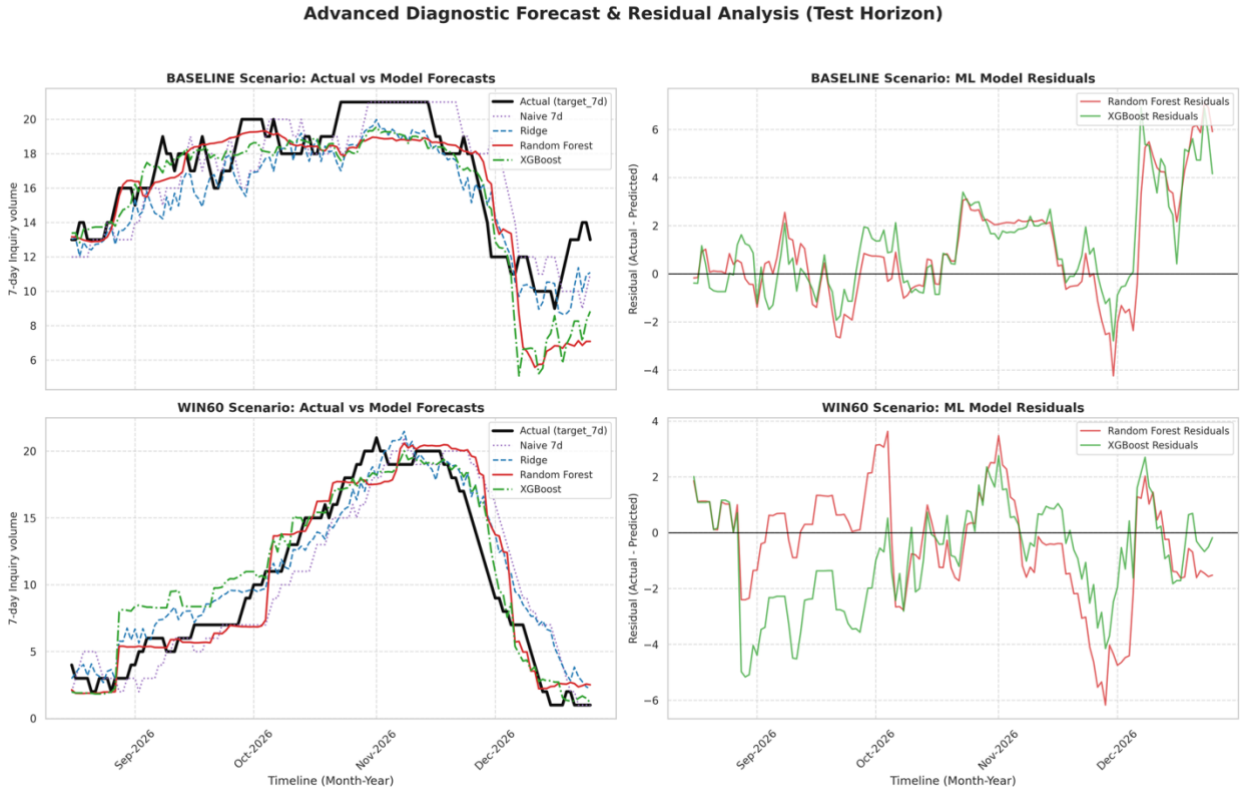
(Used to guide model selection, Google Colab prompt engineering, feature pipeline design, and the Data Science Lifecycle)

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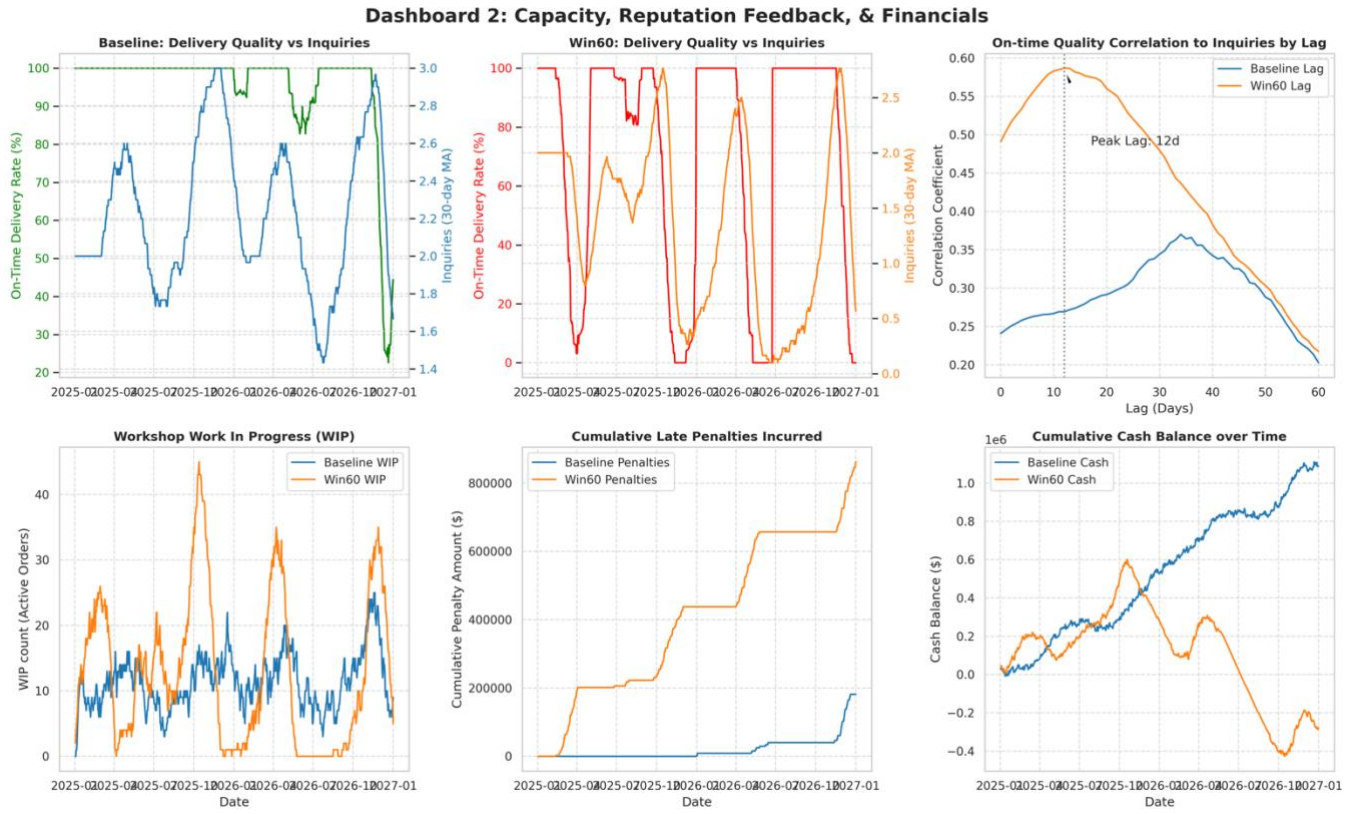
Appendix



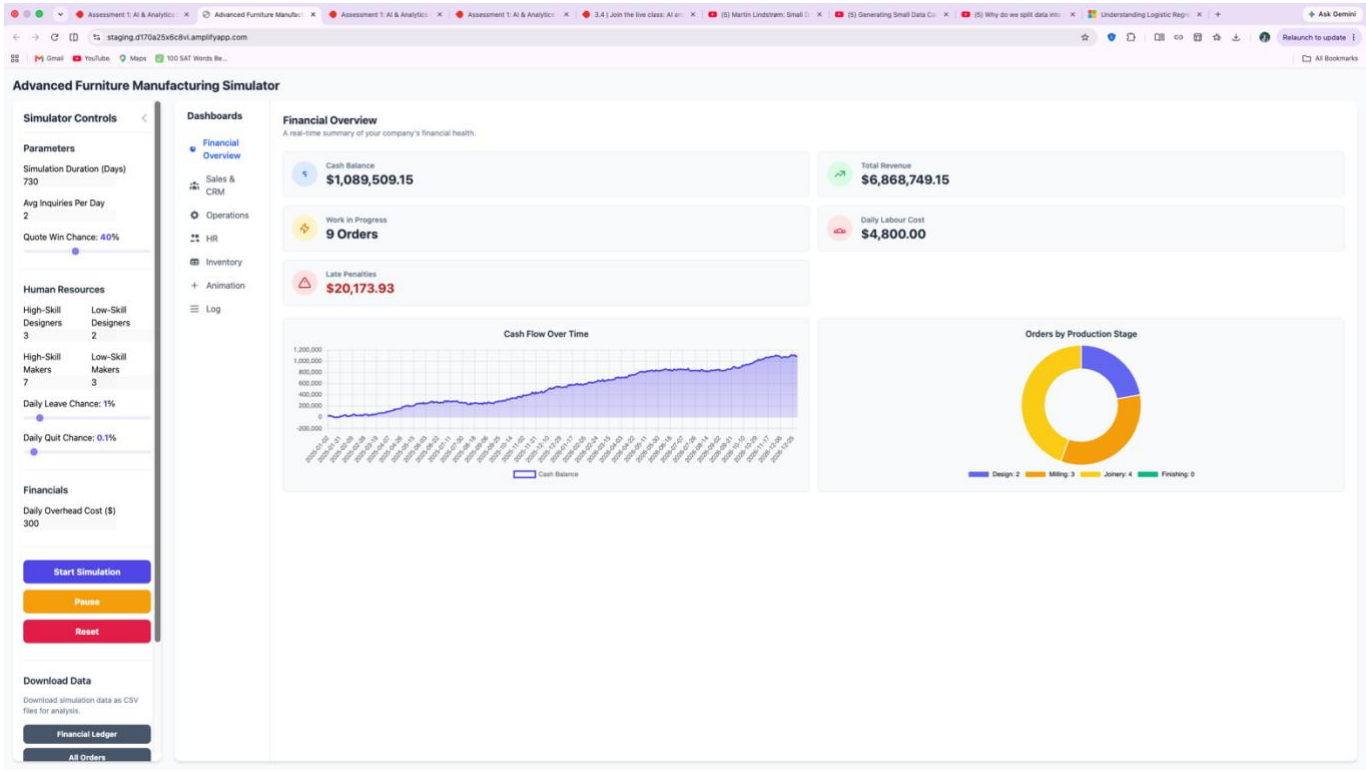
Appendix Figure A Forecasts and residuals for all models on the test period.



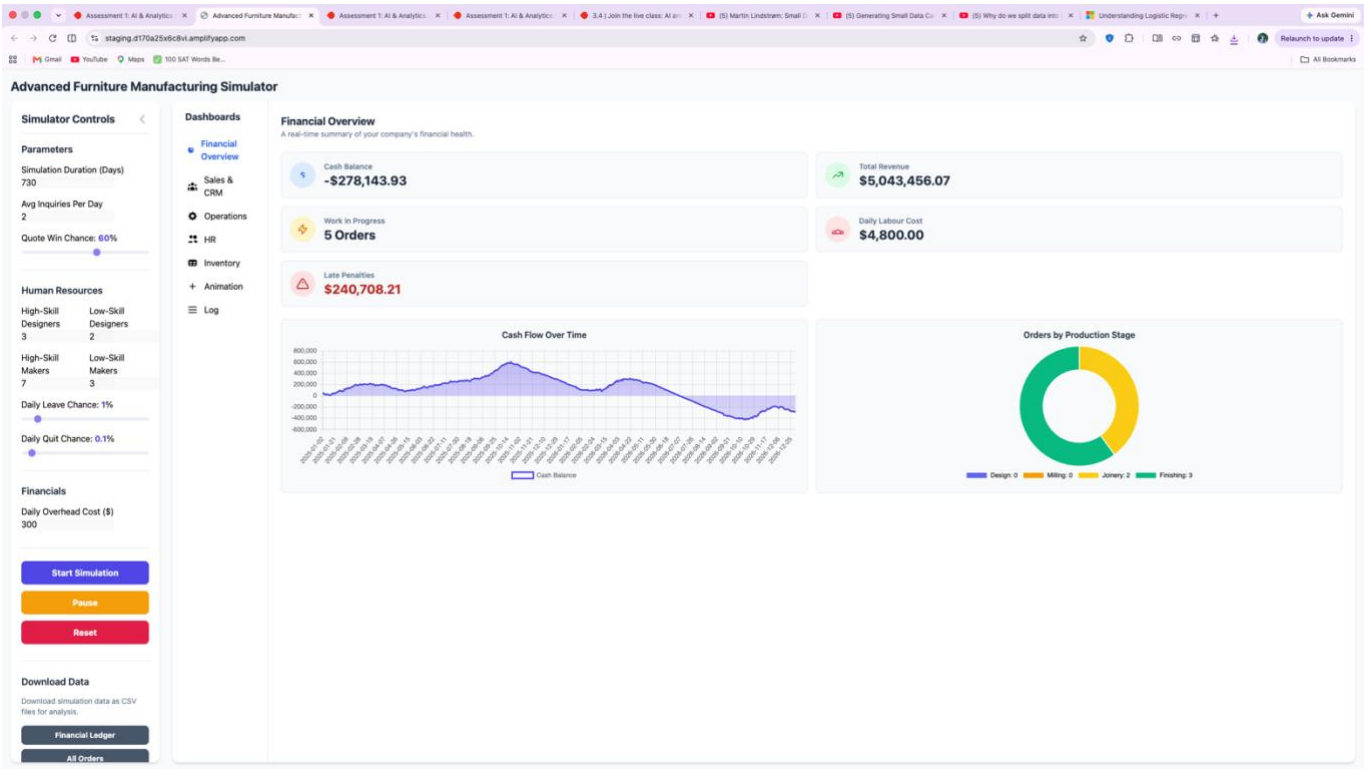
Appendix Figure B Dashboard 1- Demand Forecasting (Colab)



Appendix Figure C Dashboard 2 - Sales Optimisation (Colab)



Appendix Figure D Financials dashboard for baseline 40% simulation _ Run A



Appendix Figure E Financials dashboard for win 60% simulation_ Run B

Prompts in Colab

- I have included the pdf file titled – 'Prompts used in Gemini for Colab' in the zip file submitted. It includes all the prompts I used and Gemini responses as well.

AI usage

- I used Notebook LM to go through the research papers and understand how ML models can be used to work on problems related to demand forecasting and sales optimisation.
- Notebook LM helped me understand how Ridge, Random Forest and XGBoost learn and optimise, which informed my model choice.
- I used Claude to refine my prompts, that I used in Gemini for Google Colab python code generation.